PRISM-XAI: Privacy-Accounted Semantic Explanation Scheduling for Smart-Grid Edge Intelligence

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***Abstract*—In smart-grid edge monitoring, communicating explanations from distributed endpoints introduces wireless transmission overhead and cumulative privacy expenditure. Scheduling these updates therefore requires balancing communication resources, explanation availability, and privacy constraints. We propose a two-layer framework comprising MAP-XAI and PRISM-XAI. MAP-XAI constructs explanation packets by clipping attribution changes, applying Gaussian perturbation, and subsequently performing Top-k selection and quantization, with cumulative privacy expenditure tracked using Rényi differential privacy. PRISM-XAI selects endpoints and packet resolutions, or abstains from transmission, using declared public-state information on channel conditions, explanation freshness, event criticality, forecast novelty, and fairness, without directly accessing private attribution values. We evaluate the framework using OPSD and UCI electricity datasets in a simulated wireless uplink environment. The primary evaluation comprises 720 runs, including 120 paired MAP-XAI-only and PRISM-XAI comparisons. Under predefined service-matching criteria and comparable recorded privacy expenditure, PRISM-XAI reduces transmitted traffic by 15.72% relative to MAP-XAI-only, with a paired mean reduction of 36.40 ± 3.32 kbit (95% confidence interval half-width). Differences in the evaluated critical-event coverage and recall proxies are not statistically detected, although their absolute levels remain low. Fixed-DP baselines operate at different service levels and are therefore not used to establish matched-service superiority. Top-K attribution overlap exceeds the random reference on OPSD but not on UCI, providing dataset-dependent agreement evidence rather than demonstrated operational explanation utility. These results indicate that public-state scheduling can reduce explanation traffic relative to the packetization-only configuration under the tested conditions. Privacy accounting applies to released packet contents under the stated assumptions and does not establish privacy protection for transmission timing, endpoint selection, or the complete communication transcript.**

***Index Terms*—smart-grid communications, load-forecast explanations, explainable AI, differential privacy, explanation scheduling, resource allocation, Age of Explanation**

I. Introduction

Short-term electricity-load forecasting supports grid monitoring, demand management, and operational planning. Although machine-learning models can provide accurate forecasts, operators may also need to understand which inputs influence individual predictions. Explainable artificial intelligence (XAI) methods, such as SHAP [3] and Integrated Gradients [4], help identify the contributions of lagged demand, rolling statistics, and calendar features. Recent research has examined XAI for electricity load forecasting [16] and explainable anomaly detection in grid monitoring [15]. These studies demonstrate the relevance of model explanations in energy systems. However, identifying influential features does not establish the physical causes of demand changes, and generating explanations is only one part of making them available for operational use.

In distributed grid-monitoring systems, explanations generated at household endpoints may need to be communicated repeatedly to an edge server or operator. This creates three practical challenges. First, household-level explanations may reveal sensitive information about electricity consumption and user behaviour [17]. Second, transmitting detailed explanations consumes wireless resources and may introduce delays. Third, protecting each transmission does not remove the privacy cost of repeated releases: privacy expenditure accumulates over time, limiting how frequently explanations can be shared. Therefore, transmitting every explanation is neither necessarily communication-efficient nor compatible with a fixed long-term privacy budget.

Recent studies have addressed related challenges from different perspectives. Federated-XAI research identifies explanation privacy, communication overhead, and continual explainability as important challenges [6]. HXAI [23] investigates hierarchical, privacy-preserving explainability in distributed energy systems, motivating the need to protect explanation information shared across system levels. Related privacy-adaptive learning approaches examine the balance between privacy and utility [7], [8]. Meanwhile, task-oriented communication prioritizes information according to downstream relevance [19], while recent information-freshness research studies update scheduling under limited transmission opportunities [21], [22]. However, these research directions do not directly resolve how to schedule repeated attribution-packet releases while jointly accounting for wireless resources, explanation fidelity, operational deadlines, and cumulative household privacy expenditure.

The remaining gap is the joint design of private explanation packets and their wireless delivery. Unlike privacy-aware explanation generation such as HXAI [23], repeated transmission must balance fidelity against a finite privacy allowance: frequent releases increase coverage but may demand more noise per release. This motivates deciding *what to send, when, and when to abstain* under privacy, wireless, and operational constraints. In the considered edge-monitoring architecture, household endpoints share the uplink used to update an operator's view of forecast behaviour. An explanation

update differs from conventional fixed-format telemetry: its semantic resolution determines packet size, its value decays while the operator holds an older explanation, and each private release consumes part of a finite horizon budget. Consequently, communication control cannot be reduced to choosing the endpoint with the strongest channel. It must choose both the endpoint and explanation resolution while enforcing resource-block capacity, fading-dependent latency, delivery deadlines, freshness, and privacy admission. PRISM-XAI addresses precisely this cross-layer scheduling problem.

To address this gap, we contribute a two-layer framework. MAP-XAI is the privacy-accounted representation layer: it clips attribution changes, adds Gaussian noise, compresses the resulting private vector, and admits releases through cumulative RDP accounting. PRISM-XAI is the resource-allocation layer: it uses AoX, public grid criticality, channel state, endpoint fairness, and a public forecast-novelty score to choose endpoints, modes, or abstention. Thus MAP-XAI and PRISM-XAI are complementary contributions, and the MAP-XAI-only configuration provides the component ablation. Specifically:

- **Two-layer co-design:** MAP-XAI provides private selective explanation packets; PRISM-XAI schedules them without consulting the raw attribution-residual norm.
- **Explanation-aware uplink scheduling:** the scheduler jointly selects household endpoints, semantic packet resolution, and abstention under resource-block, deadline, channel, freshness, and privacy constraints arising in shared smart-grid edge links.
- **Matched component evidence:** over 120 paired runs, PRISM-XAI lowers traffic by 15.72% relative to MAP-XAI alone while preserving the tested coverage, recall, fairness, and privacy expenditure.
- **Robust evaluation:** 2,160 experiments across two datasets, two explanation methods, three privacy budgets, scheduler variants, and paired random seeds quantify traffic–service trade-offs.

Our objective is to characterize when selective, privacy-accounted explanation delivery is feasible and what trade-offs it introduces, rather than to assume that reducing communication alone improves operational performance.

The rest of the paper keeps the two contributions separate. Section II first defines MAP-XAI's packet mechanism and privacy accountant, and then describes the PRISM-XAI scheduler that operates on the available packet modes. Section III explains the datasets, service metrics, matched-comparison rule, and statistical protocol before presenting the main and sensitivity results. We end by separating what the experiment establishes from the stronger privacy and operational claims that remain future work.

II. Two-Layer Private Explanation Delivery

A. System and threat model

Let $\mathcal{H} = \{1, \dots, N\}$ denote household endpoints grouped into zones $\mathcal{H}_z$. At slot t, endpoint i observes $\boldsymbol{x}_{i,t} \in \mathbb{R}^d$, a deployed forecaster f produces $\hat{y}_{i,t}$, and an explainer produces $\boldsymbol{\phi}_{i,t} = \mathcal{E}(f, \boldsymbol{x}_{i,t}) \in \mathbb{R}^d$. Raw features and non-private attributions remain local. The trusted endpoint privatizes a packet before exposing it to the edge; the edge and operator may observe all transmitted packet contents and scheduling outcomes. Timing side-channels are outside the formal content-DP guarantee and require fixed opportunities or dummy traffic in timing-sensitive deployments.

TABLE I
Separation of responsibilities in the proposed stack.

Layer	Inputs used	Output and responsibility
MAP-XAI	Local attribution change; clipping radius; noise, Top-k, and quantization settings	Private packet alternatives and their RDP costs
PRISM-XAI	Public forecast novelty, criticality, AoX, channel state, fairness state, and packet metadata	Endpoint, packet mode, or abstention under the slot budget
Accountant	Per-release noise multiplier and previous admitted releases	Admit/reject decision and cumulative privacy expenditure

Figure 1 can be read from left to right. First, a household computes a forecast and a local attribution vector. Second, MAP-XAI converts the attribution change into one or more privacy-accounted packet options. Third, PRISM-XAI sees only declared public state and the metadata of those options; it chooses one option or abstains. Finally, the edge reconstructs the received private explanation. MAP-XAI therefore answers "how can an explanation be released?", whereas PRISM-XAI answers "which admissible release should use the current wireless opportunity?"

B. MAP-XAI: privacy-protected packetization

Let t_i^- denote the last explanation successfully delivered to the operator before slot t. The endpoint forms $\Delta\boldsymbol{\phi}_{i,t} = \boldsymbol{\phi}_{i,t} - \boldsymbol{\phi}_{i,t_i^-}$ and clips it to ℓ_2 radius C_i:

$$\overline{\Delta\boldsymbol{\phi}}_{i,t} = \frac{\Delta\boldsymbol{\phi}_{i,t}}{\max\{1, \|\Delta\boldsymbol{\phi}_{i,t}\|_2 / C_i\}}. \tag{1}$$

For an add/remove adjacency in which an absent household contributes the zero vector, a clipped contribution has sensitivity at most C_i. This bound applies only if the reference attribution is public, fixed, or computed entirely from previously released private outputs; otherwise its dependence on private data must also be included in the sensitivity analysis. A Gaussian mechanism then releases

$$\widetilde{\Delta\boldsymbol{\phi}}_{i,t} = \overline{\Delta\boldsymbol{\phi}}_{i,t} + \boldsymbol{n}_{i,t}, \quad \boldsymbol{n}_{i,t} \sim \mathcal{N}(0, \sigma_{i,t}^2 \Delta_i^2 \boldsymbol{I}_d). \tag{2}$$

Only after Eq. (2) does MAP-XAI retain Top-k coordinates and quantize their magnitudes. Hence selection and quantization are DP post-processing. A mode $m = (k, q, \sigma)$ has packet size

$$S_i(m) = S_0 + k(\lceil \log_2 d \rceil + q + 1), \tag{3}$$

where the final bit carries the sign. Different k, q, and σ values define packet-resolution/privacy levels; a no-release action spends no privacy. We account for repeated release with Rényi-DP (RDP) [2]. At order α, a Gaussian release costs $\rho_{i,t}(\alpha) = \alpha/(2\sigma_{i,t}^2)$. The endpoint composes these costs

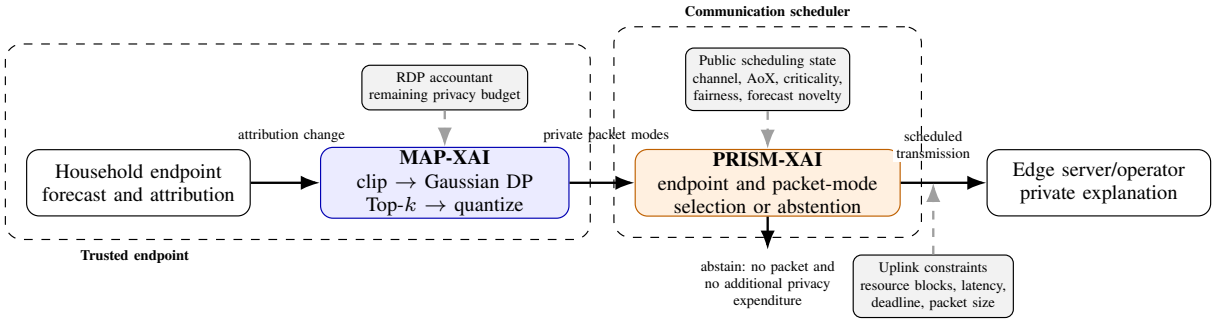


Fig. 1. PRISM-XAI communication architecture. MAP-XAI forms privacy-accounted explanation-packet modes locally. PRISM-XAI uses public grid and channel state to select an endpoint and packet resolution, or to abstain, subject to uplink resource and deadline constraints.

and rejects a private release that would exceed the remaining budget. At the end of the protected horizon,

$$\epsilon_i(\delta_i) = \min_{\alpha} \left\{ \sum_t a_{i,t}\rho_{i,t}(\alpha) + \frac{\log(1/\delta_i)}{\alpha - 1} \right\}. \tag{4}$$

Privacy is charged when a private packet is exposed outside the trusted endpoint, even if it later misses the communication deadline. The accountant must use the actual noise multiplier for every exposed release. Because clipping fixes the sensitivity, Eq. (2) is Gaussian DP at each release; Top-k selection, quantization, and packet reconstruction are post-processing. RDP composition in Eq. (4) therefore provides the stated user-level horizon guarantee whenever every admitted release remains inside the tracked RDP budget.

a) Content-privacy guarantee.: Under the stipulated add/remove contribution adjacency (not arbitrary replacement of an endpoint's entire raw history), each exposed noisy delta has order-α RDP cost $\rho_{i,t}(\alpha) = \alpha/(2\sigma_{i,t}^2)$. Adaptive RDP composition is additive for mechanisms satisfying the stated conditional sensitivity and noise assumptions. Subject to this conditional sensitivity bound and correct accounting, the resulting packet-content transcript satisfies $(\epsilon_i(\delta_i), \delta_i)$-DP if the accountant enforces the chosen horizon budget at every release. Top-k selection, quantization, reconstruction, and controller operations that depend only on public state and previously released private outputs are post-processing. This guarantee is for released content; an independent timing side channel remains outside the threat model stated above.

C. PRISM-XAI: privacy-accounted scheduling

Let $A_{i,t}$ be AoX and $w_{i,t}^{\text{crit}} \geq 1$ a public criticality weight. A candidate mode must fit the available resource blocks, meet the $D_i^{\max}$ deadline, and pass RDP admission. The controller estimates $G_{i,t}(m)$, the expected reduction in explanation loss, from AoX, public criticality, channel state, mode resolution, remaining budget, and forecast novelty; it never uses the raw attribution residual or a non-private explanation oracle for candidate selection. Forecast novelty is scaled by the calibrated clipping norm and the candidate DP-noise floor, so a private release is considered only when this declared public score is large enough relative to the expected perturbation. For a feasible mode, define gain per transmitted bit

$$\Gamma_{i,t}(m) = \frac{G_{i,t}(m)}{S_i(m)}. \tag{5}$$

The nonzero candidate set includes a MAP-XAI packet mode, optional higher-resolution modes, and a coarse zonal mode; zero transmission is always available. A richer mode replaces the MAP-XAI mode only when its gain per bit exceeds the MAP-XAI value by a configured margin. The slot utility also rewards criticality-weighted AoX and under-served endpoints while penalizing energy and RDP shadow price. The resulting multiple-choice knapsack selects at most one mode per endpoint under the slot resource-block limit. A minimum critical-event coverage target and minimum Jain fairness are evaluated as service guardrails; their reported satisfaction is empirical rather than a universal guarantee. This structure makes MAP-XAI the privacy-accounted packetization layer and PRISM-XAI the resource-allocation layer above it. With M candidate modes per endpoint and B available resource blocks, the slot-level multiple-choice knapsack can be solved by dynamic programming in $O(NMB)$ time; in our experiments M and B are small constants. The stated $O(NMB)$ bound covers only the integer-budget knapsack core, not gain estimation, mode feasibility, or cross-slot state updates. The online decision does not require a non-private oracle at the edge.

a) Slot-level operation.: At the beginning of a slot, every endpoint updates its AoX and the controller forms public criticality and channel features. MAP-XAI enumerates packet modes whose projected RDP charge fits the endpoint's remaining budget. PRISM-XAI removes modes that cannot meet the deadline or resource-block constraint, scores the surviving modes using Eq. (5), and solves the multiple-choice knapsack. An endpoint with no selected mode remains silent and spends no additional content-privacy budget. For a selected endpoint, privacy is charged when the protected packet is exposed outside the trusted endpoint, not only when delivery succeeds. The receiver updates its stored explanation and resets AoX only after successful delivery. This ordering prevents a failed radio transmission from being treated as either a free privacy trial or a successful freshness update.

b) Transmission Abstention Mechanism.: The scheduler is not required to send a low-value packet merely because a

resource block is idle. A release can be infeasible because its privacy charge exceeds the remaining allowance, because its expected latency misses the deadline, or because its public expected gain is too small relative to packet size. In each case abstention preserves resources for later slots. This is the main conceptual difference from a periodic private sender; its benefit is evaluated only against MAP-XAI-only at a matched service point.

III. Evaluation Protocol, Result, and Discussion

A. Data, wireless setting, and baselines

We use Open Power System Data (OPSD) household data and UCI ElectricityLoadDiagrams20112014 [9], [10]. Nine OPSD endpoints are grouped into three synthetic zones and 32 UCI endpoints into four synthetic zones. Each forecaster uses 12 lag, rolling-statistic, and calendar features. The tree model obtains RMSE/MAE $0.926/0.545$ on OPSD and $1.289/0.221$ on UCI. We evaluate two attribution implementations, Tree-SHAP and feature ablation, to ensure the scheduler conclusions can be examined across two attribution implementations. The uplink abstraction uses eight 180 kHz resource blocks, 1 ms slots, 5 ms deadlines, 3.5 GHz carrier frequency, path loss, correlated shadowing, small-scale fading, edge delay, and bursty failures inspired by 3GPP TR 38.901 [11]. It is a reproducible scheduler-level stress trace, not a link-level field validation. For the operational proxy, a zone is critical if its public predicted load exceeds its zone-specific calibration 80th percentile or its absolute forecast ramp exceeds the 90th percentile. A critical event is covered when zone-mean AoX is at most eight slots and its mean public quality certificate is at least 0.10. The dominant explanation driver is the largest absolute-attribution sum across four fixed feature groups; critical-event recall counts correctly identified drivers over all critical events. These are simulated decision proxies, not verified physical grid interventions.

Each matrix uses 512 evaluation slots after 96 calibration slots, $\epsilon_{\max} \in \{8, 16, 32\}$, $\delta = 10^{-5}$, a 5% coverage target, and 0.88 minimum Jain fairness. The primary public-score M8 matrix contains 2 datasets $\times$ 2 XAI methods $\times$ 3 budgets $\times$ 6 schedulers $\times$ 10 paired seeds $=$ 720 runs. We repeat the full matrix for a residual-score M8 variant and an M4 release-cap sensitivity, yielding 2,160 experiments in total. External baselines are periodic fixed-DP, channel-aware fixed-DP, AoX-aware fixed-DP, and joint scheduling with full, uncompressed private packets. We additionally report *MAP-XAI-only*, which is not an external baseline: it restricts our stack to the MAP-XAI packet mode and therefore isolates the incremental effect of full PRISM scheduling. Metrics include transmitted traffic, Top-K and cosine agreement, critical-event coverage and recall, intervention precision, false intervention, bits per correct intervention, deadline misses, Jain fairness, and spent privacy.

Coverage and recall are deliberately distinct. Coverage asks whether a usable explanation was available when a critical event occurred. Recall is stricter: an event counts only when the explanation was available and its dominant feature group agreed with the reference. A method can therefore save traffic by abstaining, but that choice remains visible in both service metrics. Top-K overlap is a fidelity diagnostic, not proof that an explanation helps an operator make a better decision.

TABLE II
Experimental protocol and interpretation of the primary metrics.

Item	Definition used in the evaluation
Unit of pairing	Same dataset, XAI method, privacy budget, and random seed
Primary contrast	Public-score PRISM-XAI minus MAP-XAI-only
Traffic	Total transmitted packet bits divided by 10^3
Coverage	Fraction of critical events meeting both AoX and public-quality thresholds
Critical recall	Correctly identified dominant drivers divided by all critical events
Top-K fidelity	Overlap between private and non-private selected features
Matched region	Coverage and recall gaps ≤ 0.02; relative spent-ϵ gap $\leq 5\%$
Multiplicity	Benjamini–Hochberg correction within each comparison family

B. Statistical analysis and Feasibility diagnostics

All headline differences use paired seeds so that both methods experience the same dataset, attribution implementation, privacy budget, and stochastic trace. We report the paired mean difference with a 95% confidence-interval half-width and compute both a paired t-test and a Wilcoxon signed-rank test. Benjamini–Hochberg adjustment is applied within each declared comparison family. Statistical significance does not replace the matched-region rule: a traffic comparison is called matched only when service and recorded privacy expenditure also satisfy Table II. Three scheduler configurations isolate the source of the traffic reduction: residual-score scheduling with $M = 8$, the same score with a stricter $M = 4$ release cap, and public-score scheduling with $M = 8$. The last configuration is the primary method; the other two are sensitivity analyses.

A separate three-seed feasibility ladder at $\epsilon_{\max} = 4$ removes one transformation at a time: (A) oracle explanation, (B) clipping only, (C) clipping plus cumulative Gaussian DP with a full packet, and (D) DP plus attribution compression. A release sweep then varies $\epsilon \in \{2, 4, 8, 16, 32\}$ and the maximum private releases per endpoint $M \in \{4, 8, 16, 32, 64\}$. These diagnostics are not used to tune the reported 720-run matrix; they identify the mechanism that limits explanation utility.

C. Results

1) MAP-XAI and PRISM-XAI Comparison: MAP-XAI and PRISM-XAI are the two proposed layers, not competing external methods. We therefore use MAP-XAI-only to measure the incremental effect of resource-aware endpoint/mode selection. It is the only comparator satisfying our declared matched-region rule in all 12 dataset/XAI/privacy conditions: absolute coverage and recall gaps at most 0.02 and relative actual-ϵ gap at most 5%. The public-score PRISM controller transmits 207.31 kbit on average versus 243.71 kbit for MAP-XAI-only. Across 120 seed-matched runs the difference is -36.40 ± 3.32

TABLE III
PRIMARY MAP-XAI-ONLY AND PRISM-XAI RESULTS BY DATASET (60 RUNS PER ROW).

Data	Method	kbit	Top-K	Coverage	Recall	$\epsilon_{\max}$
OPSD	MAP-only	106.15	.302	.0708	.0330	18.21
OPSD	PRISM	**87.98**	.294	.0710	.0324	18.21
UCI	MAP-only	381.26	.229	.0851	.0417	18.21
UCI	PRISM	**326.63**	.245	.0848	.0407	18.21

TABLE IV
COMMUNICATION OVERHEAD, EXPLANATION AGREEMENT, AND SERVICE METRICS ACROSS EVALUATED METHODS (120 RUNS PER METHOD). EXTERNAL BASELINES OPERATE AT DIFFERENT SERVICE LEVELS.

Method	Traffic (kbit)	Top-K	Coverage	Recall	$\epsilon^{\max}_{\text{spent}}$
Periodic fixed-DP	4645.2	0.252	0.383	0.178	10.62
Channel fixed-DP	4573.8	0.264	0.334	0.161	13.85
AoX fixed-DP	4643.6	0.255	0.875	0.386	10.82
Joint full packet	1355.0	0.247	0.196	0.085	18.66
MAP-XAI-only (ours)	243.7	0.265	0.078	0.037	18.21
PRISM-XAI (ours)	**207.3**	0.269	0.078	0.037	18.21

kbit (95% CI), a 15.72% reduction after Benjamini–Hochberg correction ($q < 10^{-40}$). Coverage changes by -0.01 percentage points and critical-event recall by -0.08 points; neither is statistically detected. Both spend the same recorded privacy budget. Table III disaggregates the primary comparison by dataset. The same qualitative pattern appears in both cases: PRISM-XAI sends fewer bits, while coverage, recall, and maximum spent privacy remain nearly unchanged. The absolute traffic level is higher on UCI because the experiment contains more endpoints. The table is descriptive; the paired, multiplicity-adjusted test over all 120 matched runs remains the inferential basis for the headline claim.

A nine-point public-score sweep varies the release threshold and per-endpoint cap. It exposes a communication–service frontier: PRISM-XAI traffic rises from 205.41–208.83 kbit at $M = 8$ to 574.63–584.33 kbit at $M = 32$, while coverage rises from 0.044–0.078 to 0.140–0.206. PRISM-XAI uses less traffic than MAP-XAI-only throughout, but all-cell service matching holds only at $M = 8$; Supplementary Sec. VIII provides the complete paired statistics, condition tables, and aggregate/OPSD/UCI figures.

2) External Baselines and Explanation Fidelity: Table IV shows that MAP-XAI-only transmits 82.0–94.8% fewer bits than the evaluated fixed-DP and joint full-packet baselines. However, these methods operate at different coverage and recall levels, so the reductions represent communication-service trade-offs rather than equivalent-service improvements. PRISM-XAI further reduces transmission overhead through adaptive scheduling. Explanation agreement remains dataset-dependent. On OPSD, Top-K overlap exceeds the random reference, whereas no such improvement is detected on UCI.

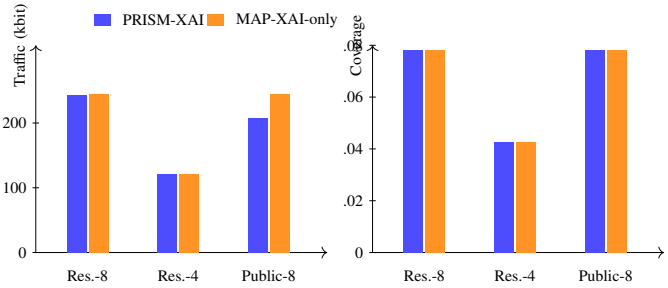


Fig. 2. MAP-XAI-only and PRISM-XAI across residual-score M8, residual-score M4, and public-score M8 configurations. Public-score M8 provides the primary result: lower traffic at essentially unchanged service.

TABLE V
PAIRED PRISM-XAI MINUS MAP-XAI-ONLY SCHEDULER SENSITIVITY ($n = 120$ PER CONFIGURATION). COVERAGE AND RECALL CHANGES ARE PERCENTAGE POINTS.

Score	M	Δ traffic (kbit)	Relative	Δ coverage	Δ recall
Residual	8	-1.19 ± 0.41	-0.50%	$+0.021$	-0.039
Residual	4	-0.02 ± 0.03	-0.00%	$+0.002$	-0.136
Public	8	$\mathbf{-36.40 \pm 3.32}$	$\mathbf{-15.72\%}$	-0.010	-0.083

These results characterize attribution agreement but do not establish operational explanation usefulness.

3) Interpretation of the Main Results: The primary comparison isolates the additional contribution of PRISM-XAI scheduling over MAP-XAI explanation compression. Across 120 paired runs, PRISM-XAI reduces traffic from 243.71 to 207.31 kbit, corresponding to a 15.72% reduction (36.40 ± 3.32 kbit, 95% confidence interval half-width). No statistically detectable differences are observed in critical-event coverage or recall, and recorded privacy expenditure is unchanged. However, the absolute coverage and recall levels remain low, so the result supports improved communication efficiency at the evaluated service level rather than deployment-ready monitoring performance. Scheduler sensitivity analysis shows that public-state scheduling produces greater traffic savings than residual-score scheduling or stricter release limits, indicating that the choice of scheduling strategy contributes to the observed improvement.

D. Discussion and Future Directions

The **main findings** demonstrate the complementary benefits of MAP-XAI and PRISM-XAI. MAP-XAI reduces transmitted traffic by 82.0–94.8% relative to periodic fixed-DP, channel-aware fixed-DP, AoX-aware fixed-DP, and joint full-packet scheduling through privacy-accounted explanation compression. These reductions reflect different communication–service operating points. Building on MAP-XAI, PRISM-XAI achieves an additional 15.72% reduction in transmitted traffic through adaptive public-state scheduling, without a statistically detectable change in critical-event coverage or recall. Both configurations have the same recorded privacy expenditure in the primary comparison. Scheduler sensitivity analysis

TABLE VI
CLAIM-TO-EVIDENCE MAP FOR THE FINAL EVALUATION.

Claim	Direct evidence	Boundary
PRISM reduces traffic	Public-score M8: -36.40 ± 3.32 kbit and -15.72% versus MAP-only	Matched component ablation; 120 pairs
Service is preserved	Coverage and recall changes are not detected	Tested thresholds, horizon, and seeds only
Privacy is accounted	No observed budget overspend; identical recorded maximum spend	Packet contents under stated adjacency and public-state assumptions
Fidelity is conditional	Above-random Top-K overlap on OPSD, not on UCI	No universal explanation-utility claim
Score choice matters	Residual-score M8 gives a smaller 0.50% traffic reduction	Scheduler sensitivity result

further shows that public-state scheduling achieves greater traffic savings than residual-score scheduling or stricter release limits. Explanation agreement remains dataset-dependent, with above-random Top-K overlap on OPSD but not on UCI.

Table VI summarizes the evidence supporting these findings. The current evaluation is limited by modest critical-event coverage and recall, dataset-dependent explanation fidelity, and simulated network conditions. Privacy accounting covers released packet contents under the stated assumptions, while other observable communication information, including transmission timing and endpoint selection, remains outside its scope. Future work will investigate application-specific explanation fidelity, longer-term privacy management, protection of transmission metadata, and evaluation under realistic network conditions with grid operators.

E. Reproducibility and Code Availability

To support reproducibility, we provide the source code, experimental configurations, and evaluation scripts in a public GitHub repository. [1] The repository includes the MAP-XAI packetization and PRISM-XAI scheduling implementations, baseline configurations, and scripts for reproducing the reported metrics and comparisons. The experiments use the publicly available OPSD and UCI electricity datasets, with fixed random seeds and documented simulation settings. Instructions for data preprocessing, experiment execution, and result generation are provided in the repository.

IV. CONCLUSION

We formulated privacy-accounted forecast explanations as schedulable smart-grid uplink traffic. MAP-XAI creates selectable private packet modes, while PRISM-XAI jointly allocates endpoints and modes under radio-resource, deadline, channel, freshness, and privacy constraints. Against the matched MAP-XAI-only component, public-score PRISM reduces traffic by 15.72% without a detected change in coverage, recall, or recorded privacy spend. External fixed-DP policies occupy different service regions. The communication result is selective explanation delivery with lower uplink traffic; it is not a claim of full-transcript DP or deployment-ready grid control. Future work should protect timing and selection, pace privacy over longer horizons, and validate the scheduler in link-level and operator-in-the-loop smart-grid trials.

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